Gesture-Based Slide Navigation Using OpenCV and Hand Detection

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1. Introduction

In the era of touchless interaction, gesture recognition has emerged as an intuitive method for interfacing with digital content. This project demonstrates a gesture-based slide control system using OpenCV and the cvzone HandTrackingModule. The system allows users to navigate through a set of presentation slides using specific hand gestures, eliminating the need for a traditional input device such as a mouse or clicker.

2. Objective

The primary objective of this project is to create an interactive system that detects hand gestures via webcam and uses them to:

- Navigate forward and backward through slides.

- Draw annotations.

- Display a pointer on slides.

- Erase annotations.

- Exit the presentation mode.

3. Required Libraries and Tools

- Python

- OpenCV – for video capture and image processing.

- cvzone – for easy hand tracking using Mediapipe.

- NumPy – for numerical operations.

- OS – to handle directories and file access.

4. System Configuration

- Operating System: Windows/Linux/macOS

- Python Version: 3.x

- Webcam: Built-in or external

- IDE: Any (e.g., VS Code, PyCharm)

5. Working / Methodology

1. Camera Setup: Captures real-time video from the webcam and flips it horizontally fornatural interaction.

2. Slide Loading: Loads all images from a designated folder to be used as slides.

3. Hand Detection: Uses the HandTrackingModule from cvzone to identify the hand andrecognize finger positions.

4. Gesture Recognition:

   - Fist (all fingers down): Move to the previous slide.

   - Thumb and Pinky Up: Move to the next slide.

   - Index + Middle + Ring: Exit the program.

   - Index + Middle: Draw on the slide.

   - Index + Middle + Ring + Pinky: Erase the last annotation.

   - Index + Middle + Ring Up: Show pointer.

5. Drawing & Annotation: Captures index finger position and allows drawing over the slides.

6. Exit Condition: The program exits when a specific gesture is shown or when 'q' is pressed.

6. IDLE Code

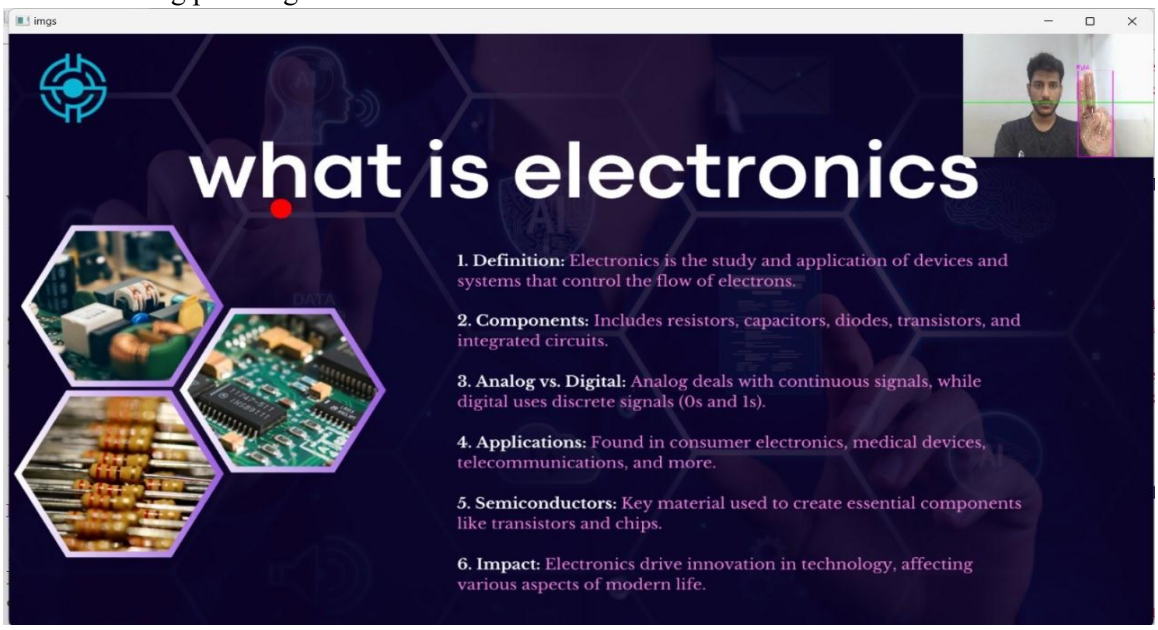
```
import cv2 import
numpy as np import os
from cvzone.HandTrackingModule import HandDetector

#variables: width,height=1280,720
folder='pres2'


# camera setup
cap=cv2.VideoCapture(0)
cap.set(3,width) cap.set(4,height)
#get list of images: imgs=os.listdir(folder)

print(imgs) #variables
imgno=6
hs,ws=int(150),int(213)
gesthreshold=400 #variables
to move slides btpress=False
btc=0 delay=20

annotations=[[]] annotationno=-1
annotationstart=False

#hand detector
detector=HandDetector(detectionCon=0.8,maxHands=1) while True:

   #import images ret,img=cap.read()
   #flipping the video horizontally
   img=cv2.flip(img,1)

   fulimg=os.path.join(folder,imgs[imgno]) imgcurr=cv2.imread(fulimg)
   imgcurr=cv2.resize(imgcurr,(width,height))

   #fliptype to name the hand right way as image is flipped
   hands,img=detector.findHands(img,flipType=False) #creating a refrence
   line to detect gesture only above that line:
   cv2.line(img,(0,gesthreshold),(width,gesthreshold),(0,255,0),5 )



   #detecting fingers:
   if hands and btpress==False:
      hand=hands[0]
      fingers=detector.fingersUp(hand)
      cx,cy=hand['center'] #print(fingers)
      lmList=hand['lmList']
```

```
#constraint value for easier drawing
xval=int(np.interp(lmList[8][0],[width//2,width],[0,width]))

yval=int(np.interp(lmList[8][1],[150, height-150],[0,height]))
indexfing=xval,yval


if cy<=gesthreshold:#if hand is at heightof face

   #gesture1:
   if fingers==[0,0,0,0,0]:
      #print('left')
      btpress=True if
      imgno>0:

         imgno-=1 annotations=[[]]
         annotationno=-1
         annotationstart=False


   #gesture2:
   if fingers==[1,0,0,0,1]:
      #print('right')
      btpress=True if
      imgno<len(imgs)-1:
      imgno+=1
         annotations=[[]] annotationno=-1
         annotationstart=False

#gesture3:showing pointer if fingers==[1,1,1,0,0]:
cv2.circle(imgcurr,indexfing,12,(0,0,255),cv2.FILLED)
#gesture4:draw if
fingers==[1,1,0,0,0]: if
annotationstart is False:
annotationstart=True
annotationno+=1
annotations.append([])
   cv2.circle(imgcurr,indexfing,12,(0,0,255),cv2.FILLED)
   annotations[annotationno].append(indexfing)
else: annotationstart=False

#gesture5:erase if
fingers==[1,1,1,1,0]: if
annotations:
annotations.pop(-1)
annotationno-=1
btpress=True
```

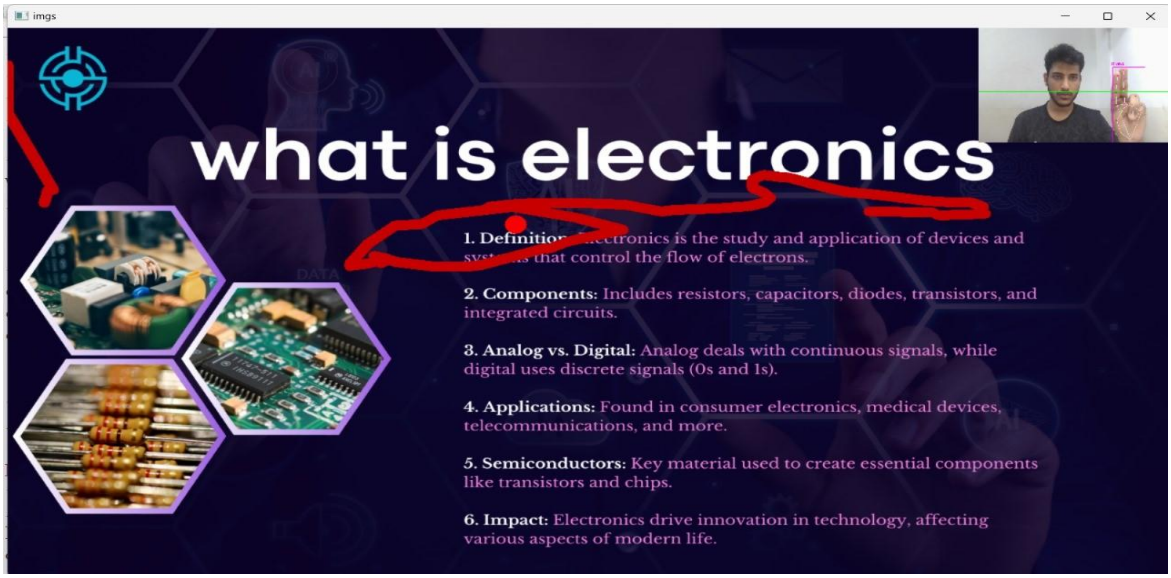
```
    #gesture6:exit if
    fingers==[0,0,1,1,1]:
    break



#setting button delay and btpress=False: if
btpress: btc+=1 if btc>delay: btpress=False
btc=0
for i in range (len(annotations)):
    for j in range (len(annotations[i])):
        if j!=0: cv2.line(imgcurr,annotations[i][j-1],annotations[i][j],(0,0,200),12)




#adding webcam img on slide
smallimg=cv2.resize(img,(ws,hs))

h,w,_=imgcurr.shape imgcurr[0:hs, w-
ws:w]=smallimg

#showing img/vid
#cv2.imshow('img',img)
cv2.imshow('imgs',imgcurr) if
cv2.waitKey(25)== ord('q'):
break
cv2.waitKey(0)
cv2.destroyAllWindows()
```

7. Output Screenshots

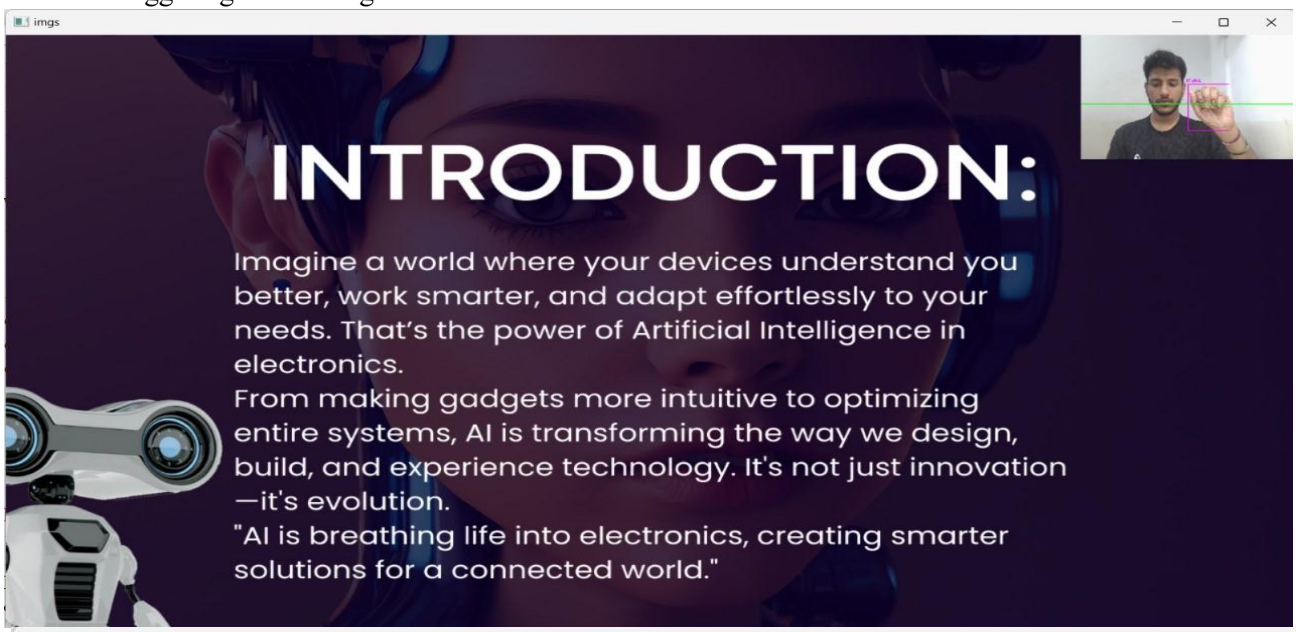
- Slide showing pointer gesture.



- Slide with drawn annotations.



- Gesture triggering slide change



8. Conclusion

This project successfully implements a contactless method to control a presentation using hand gestures. It offers a more natural and interactive experience, especially in environments where physical interaction with devices is limited. The use of computer vision techniques in such real-world applications opens the door for future enhancements like multi-hand tracking, gesture customization, and integration with actual presentation software.